



AI Rating Transparency System (ARTS)

Version 1.0/main

Authors:
Steve Sedlmayr
M Eilo

Last modified: 09/16/2026



1. Overview

1.1 Levels of Detail

ARTS consists of 4 levels of detail (LOD's). *Level of detail* is a term of art in the game industry used to describe assets viewed at different distances from the game camera. This term is indicative of the ARTS standard's cascading levels of information resolution. Note that in games, lower levels of detail actually indicate higher detail. We have inverted that, as it is more intuitive for most people to match an increasing number with increasing detail.

LOD 0: Composite AI Content Score (CACS). The lowest level of detail. This is a simple scalar value offering a rough, quantitative idea of how much AI content a product contains.

LOD 1: Rating Designator. This is an alphanumeric code, similar to an IP rating, that provides a quick, at-a-glance indication of both the amount and kind of AI content in a product, accompanied by 4 symbolic modifiers for specific, important issues, like network connectivity and system access. The alphanumeric code plus the modifiers is also referred to as a Fully Qualified Rating Designator, following the terminology for namespaces in some programming languages. The Simple Rating Designator, the alphanumeric code itself, is featured in the next LOD.

LOD 2: ARTS Label. This is a visual label that contains the Simple Rating Designator, CACS, symbolic ethics/feature modifiers, and a color code for quick visual identification. A grayscale visual bar or meter, similar to a battery or fuel indicator filling up, is provided for those with color blindness.

LOD 3: System Information Statement (SIS). The SIS is a textual document containing a full description of the product's AI content. This is intended to be a full accounting of a product's AI contents in a fully descriptive form, similar to a changelog or an SDS. Any details not caught by the ARTS Label will be covered in the SIS.

1.2 Order of coverage in this document

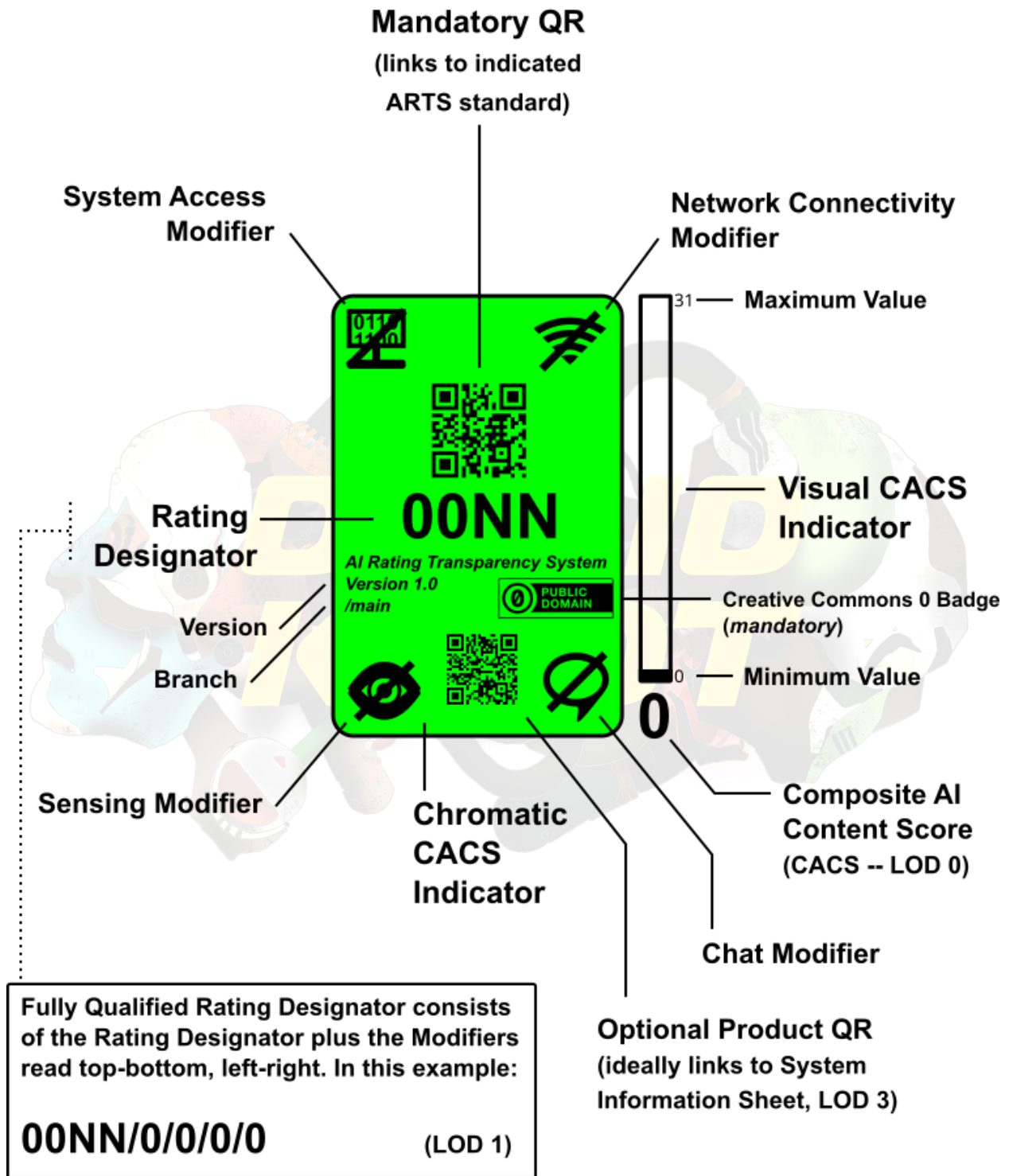
We cover the LOD's in descending order, starting from LOD 3. We chose to do this because LOD 2, the ARTS Label, is a superset of both the Rating Designator (LOD 1), and the CACS (LOD 0), which is constructed from the Rating Designator and the Modifiers.

We'll begin on the next page with a visual diagram of all of the elements of the ARTS Label, LOD 2, since it offers a diagrammatic visual overview of the entire standard.



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The AI Rating Transparency System Label (LOD 2)



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2. The ARTS Standard

2.1 LOD 3: The System Information Sheet (SIS)

The System Information Sheet is the final and highest LOD level. It is modeled after Material Safety Data Sheets under OSHA guidelines, or Safety Data Sheets under GHS. The System Information Sheet is a verbose, textual description of the precise AI specifications of the AI production, content, use and features contained in the product, presented in a technical but lay-readable format.

It will follow the following format (this definition contains terms that are explained in more detail later in the document):

1. The ARTS label
2. A textual label containing the ARTS version and branch, product name, business or individual name, CACS value and FQRD
3. A section containing a verbose description of each Position of the Rating Designator.
4. A section containing a verbose description of each Modifier as it relates to the product.
5. A section containing a verbose description of any additional specification or technical notes in list form.
6. A footer on all pages with the CC0 logo and the statement “The AI Rating Transparency System is public domain and released under a Creative Commons Zero license.” This is legally required to satisfy the CC0 license of the ARTS standard.

An example template follows on the next page (the template is available on the official github repository where this standard is also hosted).



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Statement of Information Sheet (SIS)

ARTS Version: 1.0/main

Product: XXXXX

Maker: XXXXX

LOD 0, Composite AI Content Score (CACS): 0

LOD 1, Rating Designator (RD): 00NN/0/0/0/0

LOD 2, ARTS Label

LOD 3, Statement of Information

Position 1, AI Production Scale (0): Place description here.

Position 2, Data Center Environmental Impact Score (0): Place description here.

Position 3, Production/Code Use (Type N): Place description here.

Position 4, Generative AI Use (Type N): Place description here.

System Modifiers:

System Access (0): Place description here.

Network Connectivity (0): Place description here.

Sensing (0): Place description here.

Chat Interface (0): Place description here.

Additional Specifications & Technical Notes:

Place description here.



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2.2 LOD 2: The ARTS Label

The ARTS rating label is the core of the ARTS system. It contains LOD's 1 and 0, which are derived from it, and ideally links to LOD 3, although we have left that optional in the standard (the link, not the SIS). Not optional is that every label also links to the standard itself (more detail below). The label is the visual card that will be attached to the product and clearly and conspicuously communicate to customers exactly how the product in question uses AI technology for production and deployment.

2.2.1 Rating Designator

If the rating label is the core of the ARTS system, the Rating Designator is the core of the label. The Simple RD on the card is the abbreviated version, offering an at-a-glance indicator of what the product contains, similar to an IP rating. The Fully Qualified Rating Designator, which represents LOD 1, consists of the RD from the label plus the modifiers, which we will cover in more detail in a later section that specifies LOD 1. The Simple RD consists of 4 positions, two numeric and two alphabetical, that encode the product's AI use and features disclosure.

2.2.1.1 Position 1: AI Production Scale

Position 1 discloses the scale at which AI tools were used for the production of the product. It is evaluated against a 4-point rubric:

1. How much code, if any, was generated in the production process? This can be disregarded for products that do not contain code, like videos or podcasts.
2. How much research or design was conducted using AI tools?
3. Were any models trained or deployed specifically for this product?
4. What are the security and privacy implications of this product based on how it was produced?

This rubric is evaluated in an inclusive manner, meaning if any of the statements from a scale apply to the product, then the product is rated at that scale, and is rated at the highest scale at which statements apply to the product. For instance, if a product mostly sits at scale 0, but models were specifically trained and deployed for that product, it sits at scale 2. The scales are described below.

Scale 0:

- No code was generated directly with AI.
- SLM's or LLM's may have been used to research or design systems or architecture along with traditional resources like books, Stack Overflow, forums or documentation.
- No models were trained for, or deployed along with this product.
- This product has no potential cybersecurity risks besides standard software risks that might occur from exploitable defects. Security and privacy risk is considered extremely low.



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Scale 1:

- Some code may have been generated with AI. All AI-generated code was integrated by hand and every line was reviewed and tested before integration and deployment.
- SLM's or LLM's may have been used to research or design systems or architecture along with traditional resources like books, Stack Overflow, forums or documentation.
- No models were trained for, or deployed along with this product, but models might be downloaded from the Internet to power local features.
- AI features might have file read and write access with the user's explicit permission via standard operating system dialogs or interfaces, or within the product's application domain, representing low potential cybersecurity and/or privacy risk.

Scale 2:

- A substantial amount of code was generated with AI, combined with a substantial human codebase. Integration is heavily reliant on automation, and testing of AI-generated code has 50% or less coverage of the generated portion of the codebase.
- SLM's, LLM's or agentic tools may have been used to research or design systems or architecture with or without traditional resources like books, Stack Overflow, forums or documentation.
- Models may have been trained or deployed with the product, or online, specifically for this product.
- AI features might have file read, write and delete access with or without the user's explicit permission via standard operating system dialogs or interfaces, and possibly outside the product's application domain, might expropriate files or user data to web services, and might capture limited local data via cameras or microphones aided by computer vision or other algorithms, representing a moderate to high potential cybersecurity and/or privacy risk.

Scale 3:

- Most or all of the code was generated with AI. Human interaction is mostly in the form of submitting pull requests to agentic toolsets and reviewing the output. Most of the generation is via multi-agent, long-term agentic loops. Individual software engineers may be responsible for managing dozens or hundreds of agents on a daily basis. Integration is almost entirely automated, and due to the volume of code generated on a daily basis, a very small proportion of the generated code is reviewed, and test coverage is 25% or less of the codebase.
- Most architectural design and research is also conducted via heavy reliance on agents.
- Models may have been trained or deployed with the application or online specifically for this product.
- AI features might have file read, write and delete access with or without the user's explicit permission via standard operating system dialogs or interfaces, and possibly outside the product's application domain and/or on a remote system like a source control repository, might expropriate a large number of files or user data to web services, or



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might capture large amounts of local data via cameras or microphones aided by computer vision or other algorithms, representing a high to very high potential cybersecurity and/or privacy risk.

2.2.1.2 Position 2: Environmental Impact Scale

The Environmental Impact Scale discloses a rough estimate of what the product's environmental impact is thought to be. It is divided into two parts: water impact and emissions impact. Both impacts imply usage, possible depletion of local resources, and potential pollution as possible factors in the evaluation of the score. They are both evaluated on a scale of 0-4, with a combined maximum possible score of 8. The values map roughly to Very Low, Low, Moderate, High and Very High.

Unlike Position 1, this position does not specify precise criteria for each score. The reasoning behind this is that estimating precise water and emissions impact has proven difficult to quantify and standardize in other applications. Businesses using the ARTS system are encouraged to report as accurately and honestly as possible.

2.2.1.3 Position 3: Tool Use Code

Position 3 is an alphabetic encoding of which specific kinds of tools were used in the production process. If Position 1 is considered to be a quantitative measure of this disclosure, Position 3 is the qualitative portion. Position 3 can have the following codes:

- S: SLM's (small language models)
- L: LLM's (large language models)
- A: Agentic AI
- X: Multiple of the above tools were used
- N: No AI tools were used

2.2.1.4 Position 4: Generative Art Code

Position 4 is similar to Position 3, but measures the generative art produced or used for the product. Position 4 may use one of the following codes:

- I: Images
- T: Text
- A: Audio
- V: Video
- M: 3D models or other artifacts (like the virtual 3D bounding tracks or spatial vector maps of autonomous vehicles, robots and drones)
- X: Multiple of the above were used/generated
- N: No generative artifacts were created or used for this product



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2.2.2 Modifiers

The modifiers are the 4 large icons positioned in the corners of the card. Each one calls out and discloses a specific, high-risk feature of AI systems. Read top-to-bottom and left-to-right, the 4 modifiers are:

1. **System Access Modifier:** Indicates the amount of system access a product has. These mainly concern file read, write and delete access within or outside of the product's application domain, if it is software-based, and other systems such as source control repositories or server endpoints.
2. **Network Connectivity Modifier:** Discloses how much the product communicates over a network, specifically with regards to communication with one or more AI models.
3. **Chat Modifier:** Indicates to what degree a product's interface is chat driven.
4. **Sensing Modifier:** Notifies consumers as to the degree of sensing via computer vision, sensors, cameras, microphones, telemetry, biometrics, or any other sensor-based and/or PII data potentially gathered by the product.

All of the Modifiers are rated on a scale of 0-3, representing None/Very Low, Low, Moderate and High/Very High.

2.2.3 CACS

The Composite AI Content Score, or CACS, is calculated from the combined score of the Rating Designator and the Modifiers represented as a scalar value. Standing on its own, it represents LOD 0. Specification of the CACS follows below in Section 2.4.

2.2.4 Card Color—Chromatic CACS Indicator

The card color is another quick, at-a-glance indicator of the product's score, similar to the CACS, but non-textual. It is mapped on a gradient between an RGB value of (0, 255, 0), (0.0, 1.0, 0.0) or 00FF00, i.e. primary green, at a score of 0; (255, 255, 0), (1.0, 1.0, 0.0) or FFFF00, i.e. yellow, at a score of 15; and (255, 0, 0), (1.0, 0.0, 0.0) or FF0000, i.e. primary red, at a score of 31. Fortunately, this is a clean, linear interpolation, an internationally recognized color system indicating levels of caution, and very easy to calculate precisely for each score. A table has been provided in the addenda with pre-calculated color values per score.

2.2.5 Visual CACS Indicator

For color-blind individuals or those for whom color-coding is non-intuitive, concomitant with the Chromatic CACS Indicator, i.e. the card color, ARTS also provides a visual indicator in the form of a meter or bar filling up from the minimum score to the maximum score. Note that the meter can never be fully empty, as a score of 0 is not considered a null score, but rather the first ordinal score on the CACS scale. A table has been provided in the addenda with pre-calculated height values and y-values of the visual indicator element per score.



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2.2.6 Mandatory QR Code

Every ARTS label will have a QR code affixed above the Rating Designator. This mandatory QR code will link to a place where the relevant ARTS standard is hosted:

<https://www.droidknot.com/ARTS/> in the case of Version 1.0/main, or a similar page or document in the case of an alternative version of the standard. The label is not compliant with the ARTS standard if it is missing this QR code.

2.2.7 Optional Product QR Code

The creator or business may optionally place a QR code linking to a product page, System Information Sheet or a URL where a System Information Sheet is hosted, below the Rating Designator. An SIS is mandatory for ARTS compliance, but this QR is optional. When used, it should be scaled to 75% or less of the mandatory ARTS QR code so that the mandatory QR code linking to the standard takes visual precedence.

2.2.8 Version Indicator

Displaying the version number on the label is mandatory for ARTS compliance, as different versions of the standard may have different requirements. People should have a common frame of reference and know which version of the standard is being referenced. If there are future versions of the ARTS standard, different mandatory QR codes will need to be created to point to those different standards as well.

2.2.9 Repository/Branch Indicator

For the same reason that the version number is mandatory on the label, so is the branch and/or repository of the standard. Although the ARTS standard is open-source, Droidknot and its members retain control over the main branch of the original ARTS standard on Steve Sedlmayr's personal github account, or any future repository where it might be hosted, in order to protect the stability and integrity of the standard. Anyone may fork the repository, and anyone may request to become a contributor or create a separate branch in the main repository. All requests will be reviewed and be subject to approval or denial.

If the ARTS standard is forked, the branch indicator will contain the repository and branch, like so: ***/forked_repo/branch***. If a standard resides on a permanent branch of the main repository, only the branch is required, like so: ***/branch***.

Any label not displaying this information will not be considered ARTS-compliant.

2.2.10 Creative Commons Zero Icon

An official icon for The Creative Commons Zero Universal license will be affixed to the front of all ARTS labels. This is mandatory for ARTS compliance.



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2.3 LOD 1: The Fully Qualified Rating Designator (FQRD)

As mentioned previously and illustrated in the preceding explainer diagram, the Fully Qualified Rating Designator consists of the RD from the label and the scores from all 4 modifiers, read in top-to-bottom and left-to-right order, and appended with forward slash delimiters to the end of the RD. For example: 30XN/1/2/3/4.

2.4 LOD 0: The Composite AI Content Score (CACS)

LOD 0, as previously mentioned, is the Combined AI Content Score. Calculating the CACS is straightforward. The scores from the Simple Rating Designator and Modifiers are simply added up. This breaks down as follows:

1. RD Position 1, AI Production Scale: **0-3**
2. RD Position 2, Environmental Impact Scale: **0-8**
3. RD Position 3, Tool Use Code: **0-3** (for codes of 'X', the precise number of tools used is counted)
4. RD Position 4, Generative Art Code: **0-5** (for codes of 'X', the precise number of media types used or generated is counted)
5. System Access Modifier: **0-3**
6. Network Connectivity Modifier: **0-3**
7. Sensing Modifier: **0-3**
8. Chat Modifier: **0-3**

This yields a maximum possible score of 31. 0 is not considered a null score, which means that there are 32 possible scores. Since this is a power of 2, scores can be considered to be arranged into 4 roughly exponential tiers (Tiers 3 and 4 are each extended by 1 to account for the fact that 2, 4, 8 and 16 do not sum to 32):

- **Tier 1: scores 0-1—very low or no risk**
- **Tier 2: scores 2-5—low risk**
- **Tier 3: scores 6-14—moderate risk**
- **Tier 4: scores 15-31—high risk**

Note: A Boundary Condition Regarding Platforms

We'd like to make a quick postscript about how ARTS applies to a product that resides on a proprietary platform. When creating an ARTS label, scoring it and writing a System Information Sheet for the product, it is important to separate the product from the platform on which it might be operating, as those are separate products. Otherwise, similarly to attempting to calculate the precise carbon emissions for a given product, you quickly end up in the weeds of a branching, combinatorial explosion of the entire product cycle of every ingredient and component in the product. So if you are evaluating, for instance, a YouTube channel, you do not need to include YouTube's AI rating in your own product rating. If you did so, every YouTube channel would



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have a base score of 26! A consumer, of course, may wish to consider such information. Adding YouTube's score of 26 to *Chloe VS History's* score of 16 yields a total score of 42, for instance, 11 points over the ARTS scale.



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3. Addenda

3.1 Addendum A: Color calculations¹

Red: For any CACS value of 16 or above, the value of red is 255, 1.0 or FF. At scores of 15 or below, red linearly interpolates up from 0 based on the following equation in 256-bit RGB:

$$((\text{CACS} + 1) / 16) \times 255$$

Green: For any CACS value of 15 or below, the value of green is 255, 1.0 or FF. At scores at or above 16, green linearly interpolates down towards 0 based on the following equation in 256-bit RGB:

$$((16 - ((\text{CACS} + 1) - 16)) / 16) \times 255$$

3.2 Addendum B: Visual scale calculations²

The height and y-axis value of the Visual CACS Indicator is set by the following equations:

Height: $((\text{CACS} + 1) / 32) * 294$

Y-value: $294 - \text{height} + \text{original Y-value}$

3.3 Addendum C: Score, Color and Visual Scale matrix³

CACS Score	Chromatic CACS Indicator	Visual CACS Indicator
0	10FF00	H: 9 px Y: 297
1	20FF00	H: 18 px Y: 288
2	30FF00	H: 28 px Y: 178
3	40FF00	H: 37 px Y: 269
4	50FF00	H: 46 px Y: 260

¹ All values are rounded to the nearest value.

² All values are rounded to the nearest value.

³ Y-values are based on an initial value of 12.0 in the ARTS Version 1.0/main SVG label template.



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5	60FF00	H: 55 px Y: 251
6	70FF00	H: 64 px Y: 242
7	80FF00	H: 74 px Y: 232
8	8FFF00	H: 83 px Y: 223
9	A0FF00	H: 92 px Y: 214
10	AFFF00	H: 101 px Y: 205
11	BFFF00	H: 110 px Y: 196
12	CFFF00	H: 119 px Y: 187
13	DFFF00	H: 129 px Y: 177
14	EFFF00	H: 138 px Y: 168
15	FFFF00 YELLOW	H: 147 px Y: 159
16	FFE000	H: 156 px Y: 150
17	FFD000	H: 165 px Y: 141
18	FFC000	H: 175 px Y: 131
19	FFB000	H: 184 px Y: 122
20	FFA000	H: 193 px Y: 113
21	FFA000	H: 202 px Y: 104



22	FF8F00	H: 211 px Y: 95
23	FF8000	H: 220 px Y: 86
24	FF7000	H: 230 px Y: 76
25	FF6000	H: 239 px Y: 67
26	FF5000	H: 248 px Y: 58
27	FF4000	H: 257 px Y: 49
28	FF3000	H: 266 px Y: 40
29	FF2000	H: 276 px Y: 30
30	FF1000	H: 285 px Y: 21
31	FF0000 RED	H: 294 px Y: 12

3.4 Regulatory Compliance Overview

EU AI Act Article 50 Compliance	California SB 942 Compliance	New York Synthetic Performer Law	South Korea AI Basic Act (KAIBA)
✓ 50(1), 50(3), 50(4), 50(5) ⁴ – informs and discloses the specified AI usage in a ‘clear and distinguishable manner’.	✓ Section 22757.3 – provides a conspicuous and manifest disclosure of AI content ⁵ .	✓ Section 396-b – exposes synthetic voice/art generation components explicitly.	✓ Articles 31, 33 – provides the holistic and clear product-level disclosure required, including impact level.

⁴ ARTS does not provide a machine-readable identification mark for AI-generated content, so it technically does not comply with 50(2); however, it specifies a System Information Sheet where notification of such a mark can be recorded.

⁵ ARTS does not provide a watermark or digital identification embedded in a file, and does not provide a software tool to detect AI content. Compliance with 22757.3 assumes that the provider integrates the ARTS standard into their product’s implementation.



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3.5 Legislative References

EU AI Act Article 50: <https://artificialintelligenceact.eu/article/50/>

SB-942 California AI Transparency Act (2023-2024):

https://leginfo.legislature.ca.gov/faces/billNavClient.xhtml?bill_id=202320240SB942

New York Senate Bill S8420A:

<https://www.nysenate.gov/legislation/bills/2025/S8420/amendment/A>

AI Basic Act of the Republic of Korea: <https://aibasicact.kr/explorer>

3.6 Errata

1. September 15, 2026: Corrected Section 3.4, 'Regulatory Compliance Overview', changing the row for California SB 942 to 22757.3 and amending its footnote to reflect the correct regulatory compliance for the standard. The previous table reflected a misunderstanding of the law and also listed parts of 22757.2.



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